Real Numbers

(Rational and Irrational Numbers)

02

In a class, students came up with these numbers when asked to think of all possible types of numbers.

8,	–108,	12	205
0.15,	–0.37,	0	$\frac{22}{7}$,
$\frac{3}{7}$, 1,	$\frac{10}{9}$,	$\frac{17}{19}$,	–55
–3	105	–7.5	3.2323



Does it cover all kinds of numbers?

Can you give examples of any other kind of number?

Discuss with each other and give examples.

How will we classify these numbers on the basis of their properties? Which properties do we take? Can we classified these numbers as even-odd numbers? Can they be classified by any other method? How many different ways of classifying them, can there be?

From Natural to Rational

Manisha and Salma classified these numbers as N (Natural numbers), W (Whole numbers), Z (Integers) and Q (Rational numbers) as follows-

N	W	Z	Q
8, 105, 12, 205, 1	0, 8, 105, 12, 205, 1	0, 8, 105, 12, 205, 1, –108, –55, –3	0, 8, 105, 12, 205, 1, –3, –108, –55, $\frac{22}{7}$, $\frac{3}{7}$, –0.37, 3.2323, –7.5, $\frac{10}{9}$, $\frac{17}{19}$, 0.15



You write three more numbers in all these boxes.

Are there some numbers which are there in all the boxes?

Try This



1. Write those numbers which you have in Q but not in N.
2. Write those numbes which you have in Z but not in N.
3. Write those numbers which you have in W but not in N.

All numbers that are there in box Q, can be written in the form $\frac{p}{q}$. In this p and q are integers and $q \neq 0$, that means p can be any integer and q can be any integer except zero. These are all rational numbers.

Rashmi said that natural numbers, whole numbers, Integers are also included in rational numbers because they can be written in the form $\frac{p}{q}$.

Do you agree? Reshma gave some examples like as follows-

$$8 = \frac{8}{1}, \qquad -3 = \frac{-3}{1}, \qquad 0 = \frac{0}{1}$$

Can you look for such integers which can not be written in form of rational numbers? So we can say that the group of natural numbers are included in integers and the group of integers are included in rational numbers.

Natural numbers (N) and Integers (Z) follows one rule. Each number is one more than the previous number or the number on its left side; or one less than the next number or the number on its right side. That means distance between any two consecutive numbers is always same, that is one unit.

Number Line

To show whole number on number line, draw a line and mark points at equal intervals. Assuming any one point O, write 1, 2, 3, ... at equal intervals on the right hand side.

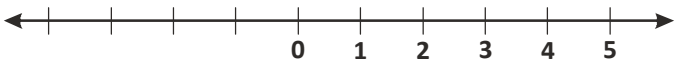


Fig. 1

Number Line for Integers

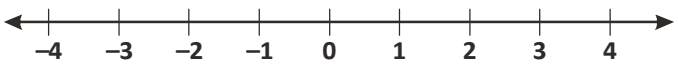


Fig. 2

To show integers on number line, write -1,-2,-3,..... on the left hand side of (*Fig.*1) number line as shown in *Fig.*2.

We can see that as we move to the right side, the number exceeds by one and as we move to the left side the number decreases by one.

Try This

1. Show on number line $-2, +3, +4$ 2. How many steps ahead +2 is form -3?



Showing Rational Numbers on Number Line.

Can you show rational number $\frac{p}{q}$ on number line?

Can you show $\frac{1}{3}, \frac{5}{3}, \frac{8}{3}$ etc. on a number line?

Rashmi showed them on number line as follows-

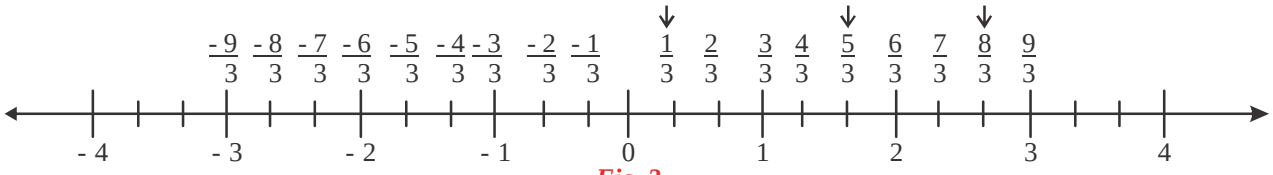


Fig. 3

In this, she divided each unit in three equal parts and then showed the numbers.

Similarly, you also try and show $\frac{7}{3}, \frac{11}{3}, -\frac{2}{3}$.

Think and discuss

What will you have to do to show $\frac{-3}{5}, \frac{2}{5}, \frac{8}{5}, \frac{12}{5}$ etc. on number line.



Equivalent Rational Numbers & Number line

Just as $\frac{1}{2}$ is rational number, similarly $\frac{2}{4}, \frac{4}{8}, \cdots\cdots$ are also rational numbers. How to we shows them on number line? Let us try and see.

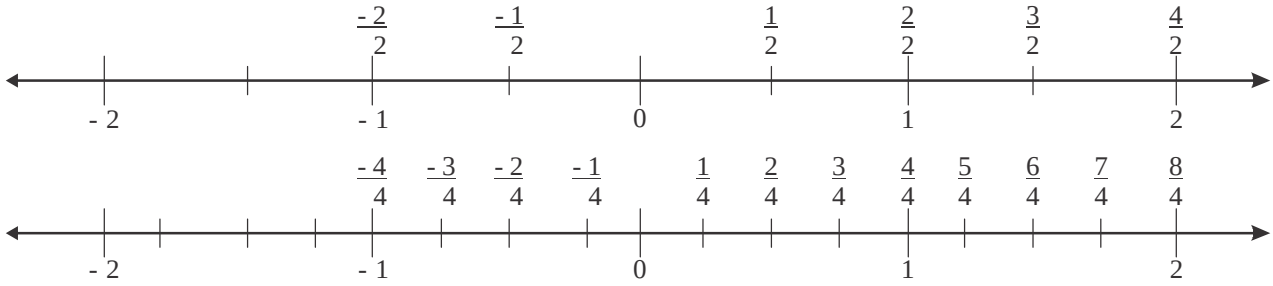


Fig. 4 (i), (ii)

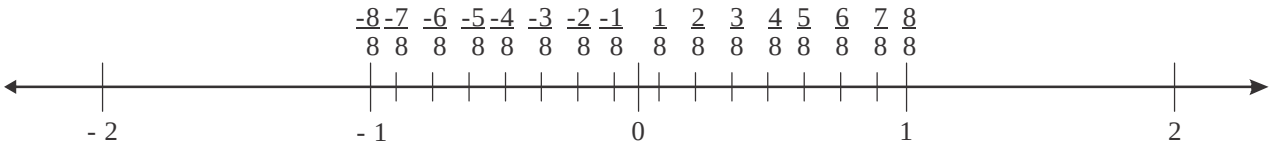


Fig. 4 (iii)

The place of $\frac{1}{2}$ on number line is also the same for $\frac{2}{4}$ and $\frac{4}{8}$. So $\frac{1}{2}, \frac{2}{4}, \frac{4}{8}$ which are equivalent rational numbers, occur at the same point on the number line.

How many rational number exist between two rational numbers– When we take any two integers, we can find the number of integers that lie between them.

Try This



1. How many integers lie between 5 & 15.
2. How many integers lie between -3 & 8.

Can we do similar counting for the rational numbers?

How many rational numbers lie between $\frac{1}{2}$ and $\frac{1}{4}$? Discuss with each other.

Reshma said– Looking at the number line, $\frac{3}{8}$ lies between $\frac{1}{2}$ and $\frac{1}{4}$.

$\frac{3}{8}$ means half of $\left(\frac{1}{2}+\frac{1}{4}\right)$

Salma said that between the rational numbers $\frac{3}{8}$ and $\frac{1}{2}$, will lie half of $\left(\frac{3}{8}+\frac{1}{2}\right)$ which means $\frac{7}{16}$. Again the mid point of $\frac{7}{16}$ and $\frac{1}{2}$ is $\frac{15}{32}$.

All these rational numbers lie between $\frac{1}{4}$ and $\frac{1}{2}$. Similarly we can find more rational numbers between these two numbers.

Can you think that how many rational numbers will lie between these two?

Discuss with each other and find some more rational numbers.

We see that as many times as we try, we can find new rational numbers lying between these rational numbers.

Now we can say that there are so many rational numbers are between $\frac{1}{4}$ *and* $\frac{1}{2}$ *that we can not count them.*

This is true for any two rational numbers.

Innumerable Numbers between Rational Numbers

a, b are any two rational number in which a < b



then $a + a < b + a$

$2a < b+a$

or $a < \frac{b+a}{2}$ ....(i)

Again $a < b$

$a + b < b + b$

$a + b < 2b$

$\frac{a+b}{2} < b$ ....(ii)

Hence $\frac{a+b}{2}$, is between a and b. i.e $a < \frac{a+b}{2} < b$

If we take any two rational numbers a and b, then there always is one rational number $\frac{a+b}{2}$ in between them and there is also a rational number between $\frac{a+b}{2}$ and a.

So we can say that there are innumerable numbers between any two rational numbers.

Finding Rational Numbers between Two Given Rational Numbers.



We take two rational number $a = \frac{5}{1}$ and $b = \frac{6}{1}$.

Although there are innumerable numbers between them, we can find some numbers using this method.

A rational number which lies between $\frac{5}{1}$ and $\frac{6}{1} = \frac{a+b}{2} = \frac{5+6}{2} = \frac{11}{2}$



$$\frac{5}{1}=\frac{5\times 2}{1\times 2}=\frac{10}{2},\ \text{(equivalent rational numbers)}$$

$$\frac{6}{1}=\frac{6\times 2}{1\times 2}=\frac{12}{2},$$

$$\frac{10}{2}<\frac{11}{2}<\frac{12}{2}$$

We can see that the rational number $\frac{11}{2}$ lies between $\frac{10}{2}$ and $\frac{12}{2}$.

Similarly

$$\frac{5}{1}=\frac{5\times 3}{1\times 3}=\frac{15}{3},$$

$$\frac{6}{1}=\frac{6\times 3}{1\times 3}=\frac{18}{3},$$

i.e. Rational numbers $\frac{16}{3},\frac{17}{3}$ lie between $\frac{15}{3}$ and $\frac{18}{3}$.

Thus, we can find several numbers between two given rational numbers by using equivalent rational numbers. For example when we multiply $\frac{5}{1}$ and $\frac{6}{1}$ by 11, we find 10 new rational numbers between these two.

If we have to find 3 rational numbers between $\frac{1}{5}$ and $\frac{2}{5}$, we will multiply the denominator and numerator of both rational numbers by one more than three i.e. 3+1 = 4.

$$\frac{1}{5}=\frac{1\times 4}{5\times 4}=\frac{4}{20}\ \text{(equivalent rational number)}$$

$$\frac{2}{5}=\frac{2\times 4}{5\times 4}=\frac{8}{20}$$

$$\frac{1}{5}=\frac{4}{20}<\frac{5}{20}<\frac{6}{20}<\frac{7}{20}<\frac{8}{20}=\frac{2}{5}$$

So three rational number $\frac{5}{20},\ \frac{6}{20},\ \frac{7}{20}$ lie between $\frac{1}{5}$ and $\frac{2}{5}$.

Try This

1. Find any 5 rational numbers between $\frac{2}{7}$ and $\frac{4}{7}$.
2. Find 3 rational numbers between $\frac{1}{5}$ and $\frac{1}{7}$.
3. Find 11 rational numbers between $\frac{-1}{3}$ and $\frac{1}{2}$.



Properties of Rational Numbers

(i) Whole Numbers and Integers

Once again discuss about properties of numbers in brief. Start with closure property.

Complete the table given below by discussion. Please give relevant example in this table.

Numbers	Operations			
	Addition	Subtraction	Multiplication	Division
Whole numbers	a + b is whole numbers for any two whole numbers a & b, which means it is closed under addition e.g.	It is not closed because 5-7 = -2 is not a whole number.	It is closed ——————— ———————	It is not closed because $5 \div 8 = \frac{5}{8}$ is not a whole number
Integers	−6+4= −2 is a integer. Integers are closed under the addition. e.g.....	It is closed because a-b is also an integer for any two integers a−b. e.g.....	5 × 9 = 45 –5× 9 = –45 and −5 x(- 9)=45 are all integers. Generally we can say that a x b is an integer for any two integers a & b. e.g.....	It is not closed because ——————— ———————

(ii) **Rational numbers - Closure Property**

(a) **Addition**



Assume two rational number $\frac{2}{7}, \frac{5}{8}$.

$$\frac{2}{7}+\frac{5}{8} = \frac{16+35}{56} = \frac{51}{56}$$

The result $\frac{51}{56}$ is again a rational number.

$8+\left(\frac{-19}{2}\right) =$ _____ Is it rational number?

$\frac{2}{7} + \frac{-2}{7} =$ _____ Will the result be a rational number?

Check this property with other numbers.

$3+\frac{5}{7}, \quad 0+\frac{1}{2}, \quad \frac{7}{2}+\frac{2}{7}$

We see that by adding any two rational numbers, the result is also a rational number. If a & b are rational numbers then $a + b$ will also be a rational number. So, rational numbers are closed with respect to addition.

(b) **Subtraction**

Assume any two rational number $\frac{5}{9}$ and $\frac{3}{4}$.

than $\frac{5}{9}-\frac{3}{4} = \frac{5\times4-3\times9}{36} = \frac{20-27}{36} = \frac{-7}{36}$

$\frac{-7}{36}$ is rational number (because -7, 36 are integers and 36 is non zero, so $\frac{-7}{36}$ also a rational number.)

Examine it with respect to the following rational numbers too.

(i) $\frac{2}{3} - \frac{3}{7} = \frac{14-9}{21} =$ _____ Is it a rational number?

(ii) $\left(\frac{48}{9}\right) - \frac{11}{18} =$ _____ Is this a rational number?

We found that for any two rational numbers, their difference is also rational number. So rational numbers are closed with respect to subtraction.

For any two rational numbers a and b, a–b is also a rational number.

(c) Multiplication



Please consider the following

(i) $3 \times \frac{1}{2} = \frac{3}{2}$

(ii) $\frac{6}{5} \times \frac{-11}{2} = \frac{-66}{10} = \frac{-33}{5}$

(iii) $\frac{3}{7} \times \frac{5}{2} =$ _____

(iv) $\frac{2}{1} \times \frac{19}{13} =$ _____

We see in all examples that the product of any two rational numbers is a rational number. Please multiply some more pairs of rational numbers. Check whether their product is a rational number or not? Can you give such rational numbers whose product is not a rational number? Therefore, it shows us that rational number are closed with respect to the multiplication.

For any two rational numbers a and b, a × b is also a rational number.

(d) Division



Take two rational numbers $\frac{2}{3}$ and $\frac{7}{8}$.

then $\frac{2}{3} \div \frac{7}{8} = \frac{2}{3} \times \frac{8}{7} = \frac{16}{21}$ which is a rational number?

Checkout it with some other examples.

(i) $\frac{5}{7} \div 2 = \frac{5}{7} \div \frac{2}{1} = \frac{5}{7} \times \frac{1}{2} = \frac{5}{14}$

(ii) $-\frac{2}{3} \div \frac{6}{11} =$ _____ = _____ = _____

(iii) $3 \div \frac{17}{13} = \frac{3}{1} \div \frac{17}{13} =$ _____ = _____ = _____

We see in all above examples that when we divide two rational numbers, the result is a rational number. Now can we say that closure property is true for the division of rational numbers?

Come, let us examine this : 0, 5 are rational numbers but $\frac{5}{0}$ is undefined. So that group of rational numbers 'Q' is not closed with respect to division.

If we remove zero from the Q than this group becomes closed with respect to division.

Try This



If we remove zero from the set of integers, is it closed with respect to division? Similarly checkout for whole numbers.

Complete the table.

Numbers	Closed under			
	Addition	Subtraction	Multiplication	Division
Natural number	Yes	.....	.....	.....
Whole numbers	.....	.....	.....	No
Integers	.....	Yes	.....	.....
Rational number	.....	.....	Yes	.....

Commutative Property

Come, Let us recall the commutative property with different operation for both whole numbers and Integers.

Complete the following table :

(i) Whole numbers

Operation	Examples	Comments
Addition	2, 3 are whole numbers 2+3 = 5 and 3 + 2 = 5	Whole numbers are commutative with respect to addition $\therefore$ 2 + 3 = 3 + 2

Subtraction	Is $3-2$ and $2-3$ the same	It is not commutative.
Multiplication	------	------
Division	$4 \div 2 = ?$ $2 \div 4 = ?$ Is $4 \div 2 = 2 \div 4$?	-------

(ii) Integers

Operations	Examples	Comments
Addition	---	Addition is commutative in integers
Subtraction	2, 3 are integers. $2-(3) = ?$ $(3)-2 = ?$ Is $2-(3) = (3)-2 = ?$	.....
Multiplication	.....	.....
Division	.....	Division is not commutative in intergers

(iii) Rational numbers

(a) **Addition**



Take rational numbers $\frac{5}{2}, \frac{-3}{4}$. Add them.

$$\frac{5}{2}+\frac{(-3)}{4} = \frac{2\times5+1\times(-3)}{4} = \frac{10-3}{4} = \frac{7}{4}$$

and $$\frac{(-3)}{4}+\frac{5}{2} = \frac{1\times(-3)+2\times5}{4} = \frac{-3+10}{4} = \frac{7}{4}$$

So, $$\frac{5}{2}+\left(\frac{-3}{4}\right) = \frac{-3}{4}+\frac{5}{2}$$

Now check this property with some more pair of rational numbers.

Take two rational numbers $\frac{1}{2}$ and $\frac{5}{7}$. Is $\frac{1}{2}+\frac{5}{7} = \frac{5}{7}+\frac{1}{2}$?

Is $$\frac{-2}{3}+\left(\frac{-4}{5}\right) = \frac{(-4)}{5}+\left(\frac{-2}{3}\right) ?$$

Can you suggest any pairs of rational numbers for which this rule is not true?

We can say that for any two rational numbers a and b, $a + b = b + a$. Thus, rational numbers are commutative under addition.

(b) **Subtraction :** Take two rational numbers $\frac{2}{3}$ and $\frac{7}{8}$.

$\frac{2}{3} - \frac{7}{8} = \frac{16-21}{24} = \frac{-5}{24}$ and $\frac{7}{8} - \frac{2}{3} = \frac{21-16}{24} = \frac{5}{24}$

So $\frac{2}{3} - \frac{7}{8} \neq \frac{7}{8} - \frac{2}{3}$

Check these

Is $2 - \frac{5}{4} = \frac{5}{4} - 2$?

Is $\frac{1}{2} - \frac{3}{5} = \frac{3}{5} - \frac{1}{2}$?



Thus, we can say that subtraction is not commutative for rational numbers.

Hence for any two rational numbers a and b, $a - b \neq b - a$.

(c) Multiplication : Take two rational numbers 2 and $\frac{-5}{7}$.

$2 \times \frac{-5}{7} = \frac{-10}{7}$; $\frac{-5}{7} \times 2 = \frac{-10}{7}$ Therefore $2 \times \frac{-5}{7} = \frac{-5}{7} \times 2$

Is $\frac{-1}{2} \times \left(\frac{-3}{4}\right) = \left(\frac{-3}{4}\right) \times \left(\frac{-1}{2}\right)$?

Check out it for some more rational numbers.

We conclude that rational numbers are commutative under multiplication.

i.e. $a \times b = b \times a$ for any two rational numbers a & b.

(d) Division



is $\frac{7}{3} \div \frac{14}{9} = \frac{14}{9} \div \frac{7}{3}$?

$\frac{7}{3} \div \frac{14}{9} = \frac{\cancel{7}^{1}}{\cancel{3}_{1}} \times \frac{\cancel{9}^{3}}{\cancel{14}_{2}} = \frac{3}{2}$ and $\frac{14}{9} \div \frac{7}{3} = \frac{\cancel{14}^{2}}{\cancel{9}_{3}} \times \frac{\cancel{3}^{1}}{\cancel{7}_{1}} = \frac{2}{3}$

$\frac{7}{3} \div \frac{14}{9} \neq \frac{14}{9} \div \frac{7}{3}$

Thus, we can say that division is not commutative for rational numbers.

Try This



Complete the table.

Numbers	Commutativity under			
	Addition	Subtraction	Multiplication	Division
Natural numbers	Yes	No	Yes	————
Whole numbers	————	————	————	No
Integers	————	————	————	————
Rational numbers	————	————	————	No

Associative Property

Recall the associativity of whole numbers relative to various operations i.e. Addition, Subtraction, Multiplication and Division.

(i) Whole numbers

Complete the table by giving necessary examples and comments.

Operation	Examples of Whole numbers	Comments
Addition	$2 + (3 + 0) = 2 + 3 = 5$ $(2 + 3) + 0 = 5 + 0 = 5$ $\Rightarrow 2 + (3 + 0) = (2 + 3) + 0$ $a + (b + c) = (a + b) + c$ For any whole numbers a, b, c	 ------- -------

Subtraction	$(4-3)-2=?\ \ 4-(3-2)=?$ Is $(4-3)-2 = 4-(3-2)$?	Subtraction is not associative
Multiplication	----------	Multiplication is associative
Division	$2\div(3\div 5)=2\div\frac{3}{5}=2\times\frac{5}{3}=\frac{10}{3}$ $(2\div 3)\div 5=\frac{2}{3}\div 5=\frac{2}{3}\times\frac{1}{5}=\frac{2}{15}$ $2\div(3\div 5)\neq(2\div 3)\div 5$	Division is not associative

(ii) Integers

Recall the associativity of integers relative to all operations.

Complete the following table with necessary comments.

Operation	Integers with examples	Comments
Addition	$2+[(-3)+5]=2+[-3+5]=2+2=4$ $[2+(-3)]+5=[2-3]+5=-1+5=4$ For any three integers a,b,c a + (b + c) = (a + b) + c	_____
Subtraction	Is 6 – (9 – 5) = (6 – 9) – 5?	_____
Multiplication	Is $2\times[7\times(-3)]=(2\times 7)\times(-3)$?	_____
Division	$10\div[2\div(-5)]=10\div\frac{-2}{5}=10\times\frac{-5}{2}=-25$ Now $(10\div 2)\div(-5)=\frac{10}{2}\div(-5)=5\div(-5)=\frac{5}{-5}=-1$ Thus $10\div[2\div(-5)]\neq(10\div 2)\div(-5)$	_____

(iii) Rational numbers - Associativity

(a) **Addition**

Assume that three rational numbers are $\frac{2}{7}$, 5, $\frac{1}{2}$. Examine if

$$\frac{2}{7}+\left[5+\left(\frac{1}{2}\right)\right]=\left[\frac{2}{7}+5\right]+\left(\frac{1}{2}\right)$$



$$\text{L.H.S.}=\frac{2}{7}+\left[5+\left(\frac{1}{2}\right)\right]=\frac{2}{7}+\left[5+\frac{1}{2}\right]=\frac{2}{7}+\left[\frac{10+1}{2}\right]=\frac{4+77}{14}=\frac{81}{14}$$

$$\text{R.H.S.}=\left[\frac{2}{7}+5\right]+\left(\frac{1}{2}\right)=\left[\left(\frac{2+35}{7}\right)\right]+\frac{1}{2}=\frac{37}{7}+\frac{1}{2}=\frac{74+7}{14}=\frac{81}{14}$$

$\therefore$ L.H.S. = R.H.S.

Find out $\frac{1}{2}+\left[\frac{3}{7}+\left(\frac{4}{3}\right)\right]$ and $\left[\frac{1}{2}+\frac{3}{7}\right]+\left(\frac{4}{3}\right)$

Are the two additions equal?

Check out the associativity of rational numbers with some more examples.

We have seen that rational numbers are associative under addition.

Therefore, for any three rational numbers a, b and c, $a + (b + c) = (a + b) + c$.

(b) **Subtraction**

Take three rational numbers $\frac{1}{2}, \frac{3}{4}$ and $\frac{-5}{4}$.

Check $\frac{1}{2}-\left[\frac{3}{4}-\left(\frac{-5}{4}\right)\right]=\left[\frac{1}{2}-\frac{3}{4}\right]-\left(\frac{-5}{4}\right)$

$$\text{L.H.S.}=\frac{1}{2}-\left[\frac{3}{4}-\left(\frac{-5}{4}\right)\right] = \frac{1}{2}-\left(\frac{3}{4}+\frac{5}{4}\right)=\frac{1}{2}-\left(\frac{8}{4}\right)$$

$$= \frac{1}{2}-2=\frac{1-4}{2}=\frac{-3}{2}$$



$$\text{R.H.S.}=\left(\frac{1}{2}-\frac{3}{4}\right)-\left(\frac{-5}{4}\right) = \left(\frac{1\times 2-3}{4}\right)+\frac{5}{4}=\left(\frac{-1}{4}\right)+\frac{5}{4}$$

$$= \frac{-1+5}{4}=\frac{4}{4}=1$$

$$\therefore \frac{1}{2}-\left[\frac{3}{4}-\left(\frac{-5}{4}\right)\right] \neq \left(\frac{1}{2}-\frac{3}{4}\right)-\left(\frac{-5}{4}\right)$$

L.H.S. $\neq$ R.H.S.

We find out that subtraction does not follow the rule of associativity in rational numbers. Therefore, for three rational numbers a, b and c ; $a-(b-c) \neq (a-b)-c$

(c) **Mulitiplication**

Take three rational numers $\frac{2}{3}, \frac{4}{7}, \frac{-5}{7}$.

Is $\frac{2}{3}\times\left[\frac{4}{7}\times\left(\frac{-5}{7}\right)\right]=\left(\frac{2}{3}\times\frac{4}{7}\right)\times\left(\frac{-5}{7}\right)$

L.H.S. = $\frac{2}{3}\times\left[\frac{4}{7}\times\left(\frac{-5}{7}\right)\right]=\frac{2}{3}\left[\frac{-20}{49}\right]=\frac{-40}{147}$

R.H.S. = $\left(\frac{2}{3}\times\frac{4}{7}\right)\times\left(\frac{-5}{7}\right)=\left(\frac{8}{21}\right)\times\left(\frac{-5}{7}\right)=\frac{-40}{147}$

L.H.S. = R.H.S.

Check this.



Find $2\times\left(\frac{1}{2}\times 3\right)$ and $\left(2\times\frac{1}{2}\right)\times 3$

Is $2\times\left(\frac{1}{2}\times 3\right)=\left(2\times\frac{1}{2}\right)\times 3$

Find $\frac{5}{3}\times\left(\frac{3}{7}\times\frac{7}{5}\right)$ and $\left(\frac{5}{3}\times\frac{3}{7}\right)\times\frac{7}{5}$

Is $\frac{5}{3}\times\left(\frac{3}{7}\times\frac{7}{5}\right)=\left(\frac{5}{3}\times\frac{3}{7}\right)\times\frac{7}{5}$

In all above situations, we get that L.H.S. = R.H.S.

Thus, multiplication is associative in rational numbers.

Hence, for any rational numbers a,b,c; $a\times(b\times c)=(a\times b)\times c$.

(d) Division

Take three rational number like– $\frac{2}{3}, \frac{3}{4}$ and $\frac{1}{7}$.

Is $\frac{2}{3} \div \left(\frac{3}{4} \div \frac{1}{7}\right) = \left(\frac{2}{3} \div \frac{3}{4}\right) \div \frac{1}{7}$?

L.H.S. $= \frac{2}{3} \div \left(\frac{3}{4} \div \frac{1}{7}\right) = \frac{2}{3} \div \left(\frac{3}{4} \times \frac{7}{1}\right) = \frac{2}{3} \div \frac{21}{4} = \frac{2}{3} \times \frac{4}{21} = \frac{8}{63}$

R.H.S. $= \left(\frac{2}{3} \div \frac{3}{4}\right) \div \frac{1}{7} = \left(\frac{2}{3} \times \frac{4}{3}\right) \div \frac{1}{7} = \left(\frac{8}{9}\right) \div \frac{1}{7} = \frac{8}{9} \times \frac{7}{1} = \frac{56}{9}$

$\frac{2}{3} \div \left(\frac{3}{4} \div \frac{1}{7}\right) \neq \left(\frac{2}{3} \div \frac{3}{4}\right) \div \frac{1}{7}$

$\therefore$ L.H.S. $\neq$ R.H.S.

Therefore, for any three rational number a, b, c ; $a \div (b \div c) \neq (a \div b) \div c$.

Thus, division is not associative in rational numbers.

Try This

Complete the table.



Numbers	Associativity under			
	Addition	Subtraction	Multiplication	Division
Natural numbers	Yes	No	.....	.....
Whole numbers	.....	.....	.....	No
Integers	.....	No	Yes	.....
Rational numbers	.....	.....	.....	.....

Role of Zero

Can you think of a number, which added to any number, gives the same number? When zero is added to any rational number, we get the same rational number.

For example–

1 + 0 = 1 and 0 + 1 = 1

–2 + 0 = –2 and 0 + (–2) = –2

$\frac{11}{3} + 0 = \frac{11}{3}$ and $0 + \frac{11}{3} = \frac{11}{3}$

Hence, we call zero as the additive identity. The statment of this property is presented below.

If a represents any rational number, then $a + 0 = a$ and $0 + a = a$

Is there an additive identity in natural numbers?

Role of 1

Fill in the blank boxes given below:

$\square \times 3 = 3$ and $3 \times \square = 3$

$-2 \times \square = -2$ and $\square \times -2 = -2$

$\frac{7}{8} \times \square = \frac{7}{8}$ and $\square \times \frac{7}{8} = \frac{7}{8}$

Have you seen some thing special in the above products?

We can see that any rational number when multiplied by '1', we get the same rational number as the product.

We can tell that '1' is the multiplicative identity for rational numbers.

For example, we do the following when we write $\frac{15}{50}$ in a simplified form.

$$\frac{15}{50} = \frac{3\times5}{10\times5} = \frac{3}{10} \times \frac{5}{5} = \frac{3}{10} \times 1 = \frac{3}{10}$$

Where we are write that $\frac{3}{10} \times 1 = \frac{3}{10}$, there we use the identity property of multiplication.

The Existence of Inverse

(i) Additive inverse

$3 + (-3) = 0$ and $-3 + 3 = 0$

$-5 + 5 = 0$ and $5 + (-5) =$ _____

$\frac{2}{3} + ? = 0$ and $? + \frac{2}{3} = 0$

$\left(-\frac{1}{2}\right) + ? = 0$ and $? + \left(-\frac{1}{2}\right) = 0$

In the first example -3 and 3 are additive inverses to each other, because we get '0' by adding those. Any two number whose sum is '0' are called additive inverses of each other. Generally If a is rational number $a + (-a) = 0$ and $(-a) + a = 0$.

Then 'a', '$-a$' are additive inverses of each other.

The additive inverse of 0, remains 0 because $0 + 0 = 0$

(ii) Multiplicative inverse

Which number should be multiplied to a rational number $\frac{2}{7}$ to give a product of 1?

We see $\frac{2}{7} \times \frac{7}{2} = 1$ and $\frac{7}{2} \times \frac{2}{7} = 1$



Fill in the following blank boxes.

$2 \times \square = 1$ and $\square \times 2 = 1$

$-5 \times \square = 1$ and $\square \times 5 = 1$

$\frac{-17}{19} \times \square = 1$ and $\square \times \frac{-17}{19} = 1$

$1 \times ? = 1$

$-1 \times ? = 1$

Any two numbers whose product is '1' are called multiplicative inverses of each other.

For example, $4 \times \frac{1}{4} = 1$ and $\frac{1}{4} \times 4 = 1$, therefore number 4 and $\frac{1}{4}$ are multiplicative inverses of each other.

We can say that if $\frac{a}{b} \times \frac{c}{d} = 1$ and $\frac{c}{d} \times \frac{a}{b} = 1$, the rational number $\frac{c}{d}$ is called multiplicative inverse of the second rational number $\frac{a}{b}$.

Rational Number and their Decimal Form

If we want to write decimal form of $\frac{7}{4}, \frac{9}{11}, \frac{7}{6}$ etc. we will divide 7 by 4, 9 by 11 and 7 by 6.

```
4 | 1.75
  | 7
  | 4
  | 30
  | 28
  | 20
  | 20
  | 0
```

```
11 | 0.8181
   | 9.0
   | 88
   | 20
   | 11
   | 90
   | 88
   | 20
   | 11
   | 9
```

```
6 | 1.1666
  | 7
  | 6
  | 10
  | 6
  | 40
  | 36
  | 40
  | 36
  | 40
  | 36
  | 4
```

Do the calculation and match with the table.

1. Remainder = 3, 2, 0 — Divisor 4
2. Remainder = 2, 9, 2, 9, 2, 9, ... — Divisor 11
3. Remainder = 1,4,4,4,4..... — Divisor 6

Look at these remainders. Can you see anything special?

1. On dividing by 4, remainder is zero after one point.
2. On dividing by 11, 2 and 9 keep recurring as reminder turn by turn.
3. On dividing by 6, the remainder is 4 repeatedly after the first time.
4. When there is a repetition of the remainder there is also a repetition observed in quotient.

To understand this, let us make a table.

Number	Quotient	Analysis	Conclusion
$\frac{7}{4}$	1.75	Remainder is zero after the two places of decimal	Terminating decimal
$\frac{9}{11}$	0.8181.....	There is a repetition of the digits after first two places of decimal	Non- terminating recurring decimal
$\frac{7}{6}$	1.166.....	There is the repetition of the number which comes after the first decimal place.	Non-terminating recurring decimal

We find from this table-

Reminder is zero after a few decimal places in number $\frac{7}{4}$. Some other example of this are $\frac{1}{2} = 0.5, \frac{7}{8} = 0.875$ etc. Such decimals are called terminating decimals, in which numbers after decimal point are limited.

The repetition occurs in the quotient after a fixed place in numbers like $\frac{9}{11}, \frac{7}{6}$. We also write $\frac{7}{6} = 1.1666...$ or as $1.1\overline{6}$. Similarly, write $\frac{9}{11} = 0.8181...$ as $0.\overline{81}$. This is non-terminating recurring i.e. unlimited and some groups of digits are repeated after some places of decimal.

Try This

1. Write two such rational numbers whose decimal form is non-terminating recurring.
2. Write two such rational numbers whose decimal form is terminating.



Writing the Decimal Form of Rational Number in its General Form

When we write decimal form of a number as form $\frac{p}{q}$ than we understood the number better. Let us take some recurring decimal numbers-

EXAMPLE-1. Express $1.555..... = 1.\overline{5}$ in form of $\frac{p}{q}$.

SOLUTION : Assume $\frac{p}{q} = 1.55555.....$ (i)

multiply both sides by 10.

$10\frac{p}{q} = 15.5555 .....$ (ii)

subtract equation (i) from (ii)

$10\frac{p}{q} - \frac{p}{q} = (15.55555 .....) - (1.555 .....)$

$9\frac{p}{q} = 14$



$$\frac{p}{q} = \frac{14}{9}$$



EXAMPLE-2. To express $7.3\overline{456}$ in the form of $\frac{p}{q}$.

SOLUTION : Assume $\frac{p}{q} = 7.3\overline{456} = 7.3456456$ .....

Multiply both sides by 10.

$$10\frac{p}{q} = 73.456456..... \quad .....(i)$$

Multiply both sides by 1000 in equation (i), we get

$$10000\frac{p}{q} = 73456.456456 ..... \quad .....(ii)$$

subtract equation (i) from equation (ii).

$$9990\ \frac{p}{q} = 73383$$

$$\frac{p}{q} = \frac{73383}{9990}$$

$$\therefore \frac{p}{q} = \frac{24461}{3330}$$

Try This



Change the following decimals in the form $\frac{p}{q}$.

(i) 1.2333....., (ii) 3.88..... (iii) 3.204343.....

Exercise-2.1



1. Give examples for the following statements.
   (i) A number which is natural number, whole number and integer.
   (ii) A number which is whole number but not the natural number.
   (iii) A number which is rational number but not the natural number.



2. Find 3 rational numbers between the 6 & 7.
3. Find 5 rational numbers between $\frac{4}{7}$ and $\frac{5}{7}$.
4. Find 3 rational numbers between $\frac{2}{3}$ and $\frac{3}{4}$.
5. Find any 4 rational numbers between -1 and 2.
6. Write any 3 rational numbers between $\frac{5}{10}$ and $\frac{9}{10}$.
7. Represent the $\frac{7}{3}$ and $-\frac{7}{3}$ on a number line.
8. Change the following rational number to decimal form and find what kind of decimal form they are?

   (i) $\frac{126}{5}$ (ii) $\frac{335}{16}$ (iii) $\frac{22}{7}$ (iv) $-\frac{118}{3}$

9. Express the following decimal numbers in $\frac{p}{q}$ form.

   (i) 0.53 (ii) 16.8 (iii) 105.25 (iv) 7.36

10. Change the following decimal numbers in $\frac{p}{q}$ form.

    (i) $0.\overline{70}$ (ii) $8.\overline{39}$ (iii) $3.12\overline{7}$ (iv) $5.\overline{125}$

Irrational Numbers

So far we have seen the decimal conversion of rational numbers, for example, $\frac{9}{18} = 0.5$, $\frac{7}{6} = 1.166... .... = 1.1\overline{6}$. It is obvious from these examples that their decimal forms are either terminating or non-terminating but recurring. Can you think of such numbers which are non-terminating but not recurring?

1.414213 .....

1.7320508 .....

2.2360679774 .....

As you can see in the above examples, neither do the decimals end nor do they repeat. These are called infinite non-recurring. See more such numbers. It can not be written in the form of $\frac{p}{q}$.

Similarly, $\sqrt{3} = 1.7320508075\ldots\ \ldots\ \ldots\ \ldots$

$\sqrt{5} = 2.2360679\ldots\ \ldots\ \ldots\ \ldots$

These too can not be written in the form of $\frac{p}{q}$. i.e. those numbers which are not the perfect squares, their root is an irrational number. It means $\sqrt{4}$, $\sqrt{36}$ is rational number because $\sqrt{4} = 2 = \frac{2}{1}$ and $\sqrt{36} = 6 = \frac{6}{1}$ but $\sqrt{3}$ and $\sqrt{7}$ is not a rational number.

Try This



Identify the rational number, irrational number in following numbers.

(i) $\sqrt{6}$ (ii) $\sqrt{7}$ (iii) $-\sqrt{25}$ (iv) $-\sqrt{8}$ (v) $\sqrt{9}$ (vi) $\sqrt{\frac{9}{16}}$

π (Pi)

$\pi = 3.14159265389.....$

π (Pi) is an irrational number which is approximated as the ratio of perimeter to the diameter of circle.

$$\pi = \frac{Perimeter}{Diameter} = \frac{c}{d}$$

Does this mean that at least one of the two numbers in the ratio is irrational or is there no such rational number which when multiplied to diameter to give us the perimeter. Often we assume the value of π as $\frac{22}{7}$ but this is just an approximate value of π.

Place Determination of Irrational Number on Number Line.

We have see that several rational numbers are found between two given rational numbers and these can be shown on the number line. Is there any place is blank between these

rational numbers on number line? If the irrational numbers can also shown on number line then we can surely say that rational numbers do not cover the whole number line. Let us see how we can determine the place of $\sqrt{2}$ on number line.

EXAMPLE-3. Locate the place of $\sqrt{2}$ on number line.

Draw a perpendicular AB on point A (*Fig.*5). Take the length of AB = 1 unit. With the help of compass draw an arc of radius OB with O as the centre. It cuts the number line at a point P. Because the value of OB is $\sqrt{2}$, so OP is also $\sqrt{2}$. It means P is the point on number line which represents $\sqrt{2}$. Hence, there can not be any rational number at this point.

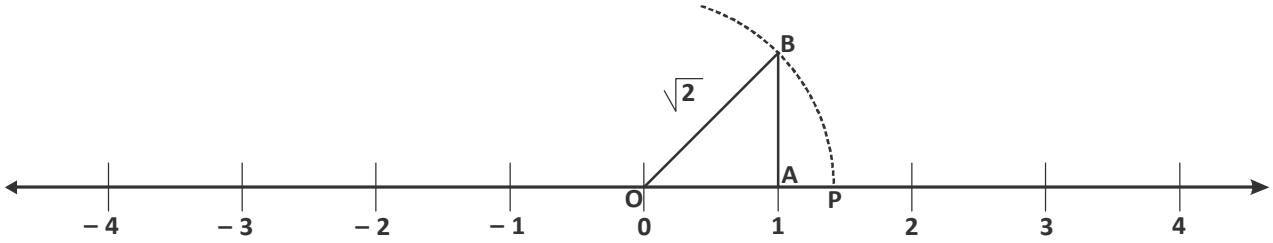


Fig. 5

Similarly, We may locate the place of $\sqrt{3}$ on number line.

Drawn a perpendicular AB at point A on number line (*Fig.*6), Where we take AB = 1 unit. Join O to B, then OB = $\sqrt{2}$.

We assume the centre 'O' and cut the arc with OB as radius.

This arc is cuts the number line at a point P. Draw a perpendicular CP at P where CP= 1 unit. Join O to C. Assume the centre O and cut the arc with OC as radius, which cuts the number line at the point Q. The distance OQ shows $\sqrt{3}$.

Point Q corresponds to $\sqrt{3}$.

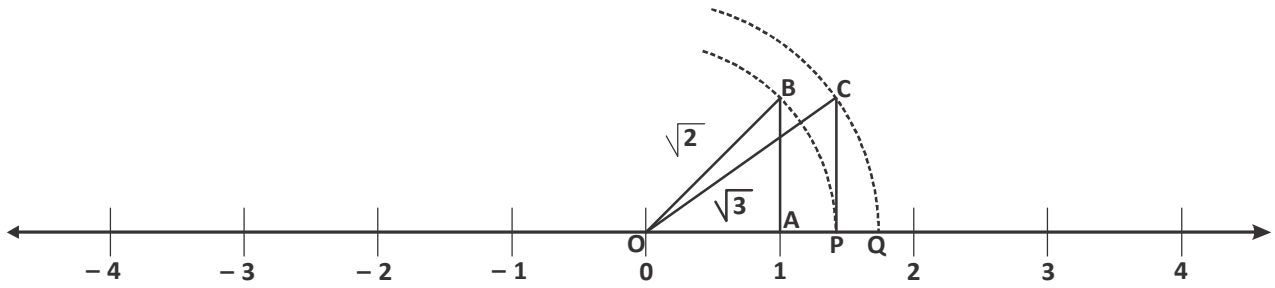


Fig. 6

Try This



1. Following numbers are rational or irrational?

   (i) $\sqrt{81}$ (ii) $-\sqrt{625}$ (iii) $\sqrt{11}$

   (iv) $-\sqrt{\frac{81}{25}}$ (v) $3.232323\ldots\ldots\ldots$ (vi) $5.7070070007\ldots$

2. Locate $\sqrt{5}, \sqrt{7}$ on number line.

3. Locate $-\sqrt{2}, -\sqrt{5}$ on number line.

Real Numbers

If we show all the rational and irrational numbers on the number line then will any number remain on number line? No, the collection which we get by combining the rational and irrational number, will cover the all points of number line. This big collection (set) is known as real numbers.

Operations on Real Numbers

We are doing operations of addition, subtraction, multiplication, division of rational numbers. We have seen that the addition and multiplication of rational number is always a rational number. Do we always get an irrational number when we do operations on irrational numbers? See the following examples–

$$\sqrt{5} + \left(-\sqrt{5}\right) = 0$$

$$\sqrt{3} \times \sqrt{3} = \sqrt{9} = 3$$

$$\frac{\sqrt{5}}{\sqrt{5}} = 1$$



In all these, the numbers which we got after operation, are not irrational. It means the addition, subtraction, multiplication & division of irrational numbers does not always give an irrational number.

Try This



1. You take some irrational numbers and check by doing operations.
2. Find atleast 5 examples where we don't get irrational number after doing some operations on them. Also think of such examples in which we get irrational number after doing operation.

Identifying Irrational Numbers

We know that $\sqrt{3}, \sqrt{5}, \sqrt{6}, \sqrt{8}, \sqrt{10}$ etc. are irrational numbers. However $\sqrt{1}, \sqrt{4}, \sqrt{9}$ are rational numbers, but are $5+\sqrt{2}$, $\sqrt{3}-1$, $\frac{\sqrt{5}}{2}$ and other such numbers irrational numbers? Let us see-

We know $\sqrt{2}$ = 1.41421....

$\sqrt{3}$ = 1.73205

$\sqrt{5}$ = 2.23606....

(i) Therefore, $5+\sqrt{2}=6.41421.....$ This representation is non terminating and non recurring. So $\left(5+\sqrt{2}\right)$ is not rational number.



(ii) $\sqrt{3}-1=\left(1.73205.....\right)-1$

$=$ 0.73205..... is non terminating and non recurring.

Therefore $\sqrt{3}-1$ is an irrational number.

(iii) $3\sqrt{2}=3\times(1.41421 \ldots \ldots \ldots \ldots)$

$=$ 4.24263 ..... is non terminating and non recurring decimal number.

Therefore $3\sqrt{2}$ is an irrational number.

(iv) $\frac{\sqrt{5}}{2}=\frac{2.23606.....}{2}$

$=$ 1.11803..... is non terminating and non recurring decimal number.

Therefore $\frac{\sqrt{5}}{2}$ is an irrational number.

What can be conclude from this?

By above examples we can say that addition, subtraction, multiplication and division between one rational and one irrational number is always irrational number.

Suppose a and b are two positive real numbers. Then

(i) $a\sqrt{b}+c\sqrt{b}=(a+c)\sqrt{b}$

(ii) $a\sqrt{b} - c\sqrt{b} = (a-c)\sqrt{b}$

(iii) $\sqrt{a}\ \sqrt{b} = \sqrt{ab}$

(iv) $\dfrac{\sqrt{a}}{\sqrt{b}} = \sqrt{\dfrac{a}{b}}$

(v) $\left(\sqrt{a}+\sqrt{b}\right)\left(\sqrt{c}+\sqrt{d}\right) = \sqrt{ac}+\sqrt{ad}+\sqrt{bc}+\sqrt{bd}$

(Where c and d are positive real numbers)

(vi) $\left(\sqrt{a}+\sqrt{b}\right)\ \left(\sqrt{a}-\sqrt{b}\right) = a-b$

(vii) $\left(a+\sqrt{b}\right)\ \left(a-\sqrt{b}\right) = a^2-b$

Try This



1. Solve these-

   (i) $3\sqrt{2}+3\sqrt{2}$ (ii) $3\sqrt{6}.2\sqrt{3}$

2. Subtract $5\sqrt{3}+7\sqrt{5}$ from $3\sqrt{5}+7\sqrt{3}$.

3. Multiply $6\sqrt{3}$ by $13\sqrt{3}$.

4. Solve these-

   (i) $\left(3+\sqrt{3}\right)\left(2+\sqrt{2}\right)$ (ii) $3\left(2+\sqrt{3}\right)\left(2-\sqrt{3}\right)$

   (iii) $\left(\sqrt{5}+\sqrt{2}\right)^2$ (iv) $\left(\sqrt{5}-\sqrt{2}\right)\left(\sqrt{5}+\sqrt{2}\right)$

Some other Operations

When there are several parts in a real number, then how do we add?

How can we add $a\sqrt{b}+c\sqrt{b}+e\sqrt{b}$? This addition $= (a+c+e)\sqrt{b}$

Similarly other real numbers can be added.

EXAMPLE-4. Add $7\sqrt{3}-5\sqrt{5}$ to $9\sqrt{3}+7\sqrt{5}$.

SOLUTION : $9\sqrt{3}+7\sqrt{5}+7\sqrt{3}-5\sqrt{5}$

$= 9\sqrt{3}+7\sqrt{3}+7\sqrt{5}-5\sqrt{5}$

$= (9+7)\sqrt{3}+(7-5)\sqrt{5}$

$= 16\sqrt{3}+2\sqrt{5}$



EXAMPLE-5. Subtract $3\sqrt{2}-2\sqrt{3}$ from $3\sqrt{3}-2\sqrt{5}$.

SOLUTION : $3\sqrt{3}-2\sqrt{5}-\left(3\sqrt{2}-2\sqrt{3}\right)$

$= 3\sqrt{3}-2\sqrt{5}-3\sqrt{2}+2\sqrt{3}$

$= 3\sqrt{3}+2\sqrt{3}-2\sqrt{5}-3\sqrt{2}$

$= (3+2)\sqrt{3}-2\sqrt{5}-3\sqrt{2}$

$= 5\sqrt{3}-2\sqrt{5}-3\sqrt{2}$

EXAMPLE-6. Solve the following-

(i) $\sqrt{3}\times\sqrt{5}$ (ii) $2\sqrt{3}\times 7\sqrt{2}$ (iii) $\left(2+\sqrt{5}\right)\left(2-\sqrt{5}\right)$

(iv) $\left(\sqrt{7}+\sqrt{3}\right)\left(\sqrt{2}+\sqrt{5}\right)$

SOLUTION : (i) $\sqrt{3}\times\sqrt{5}=\sqrt{3\times5}=\sqrt{15}$

(ii) $2\sqrt{3}\times 7\sqrt{2}=2\times7\sqrt{3\times2}=14\sqrt{6}$

(iii) $\left(2+\sqrt{5}\right)\left(2-\sqrt{5}\right)=4-5=-1$

(iv) $\left(\sqrt{7}+\sqrt{3}\right)\left(\sqrt{2}+\sqrt{5}\right)=\sqrt{14}+\sqrt{35}+\sqrt{6}+\sqrt{15}$

Rationalising the Denominator

We have shown $\sqrt{2}$ on number line, then can we show $\frac{1}{\sqrt{2}}$ on number line?

What is the value of $\frac{1}{\sqrt{2}}$.

$\frac{1}{\sqrt{2}}=\frac{1}{1.414213}....$

Making a rational number of the denominator means 'rationalisation'

Can we divide 1 by $\sqrt{2}=1.414213...$? This is not easy. Because $\sqrt{2}$ is non terminating and non recurring decimal number. In this situation we need to rationalise the denominator.

To rationalise the denominator we multiply both numerator and denominator of $\frac{1}{\sqrt{2}}$ by $\sqrt{2}$.

$$\frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{1}{2}\sqrt{2} = \frac{\sqrt{2}}{2}$$ this is half of $\sqrt{2}$.

Can we show $\frac{\sqrt{2}}{2}$ on the number line?

$\frac{\sqrt{2}}{2}$ is half of $\sqrt{2}$, This is between 0 and 1.

EXAMPLE-7. Rationalise the denominator of $\frac{1}{4-\sqrt{7}}$.

SOLUTION: $\frac{1}{4-\sqrt{7}} = \frac{1}{4-\sqrt{7}} \times \frac{4+\sqrt{7}}{4+\sqrt{7}}$ [the rationalising factor of $\left(a-\sqrt{b}\right)$ is $\left(a+\sqrt{b}\right)$]

$$= \frac{4+\sqrt{7}}{(4)^2-\left(\sqrt{7}\right)^2}$$

$$= \frac{4+\sqrt{7}}{16-7}$$

$$= \frac{4+\sqrt{7}}{9}$$



EXAMPLE-8. Rationalise the denominator of $\frac{3-\sqrt{2}}{3+\sqrt{2}}$.

SOLUTION : $\frac{3-\sqrt{2}}{3+\sqrt{2}} = \frac{3-\sqrt{2}}{3+\sqrt{2}} \times \frac{3-\sqrt{2}}{3-\sqrt{2}}$ [the rationalising factor of $\left(a+\sqrt{b}\right)$ is $\left(a-\sqrt{b}\right)$]

$$= \frac{3\times 3-3\times\sqrt{2}-\sqrt{2}\times 3+\sqrt{2}\times\sqrt{2}}{(3)^2-\left(\sqrt{2}\right)^2}$$

$$= \frac{9-6\sqrt{2}+2}{9-2}$$

$$= \frac{11-6\sqrt{2}}{7}$$

Try This



1. Rationalise the denominator.

   (i) $\frac{1}{4+\sqrt{5}}$ (ii) $\frac{1}{7+4\sqrt{3}}$ (iii) $\frac{1}{7+4\sqrt{3}}+\frac{1}{2+\sqrt{5}}$

Exercise 2.2



1. Simplify –

   (i) $7\sqrt{3}+11\sqrt{3}$ (ii) $5\sqrt{7}-2\sqrt{7}$

2. Find the sum of $2\sqrt{2}+3\sqrt{5}$ and $5\sqrt{2}-2\sqrt{5}$.

3. Subtract $3\sqrt{5}+5\sqrt{7}$ from $8\sqrt{7}-5\sqrt{5}$.

4. Simplify.

   (i) $\left(2+\sqrt{3}\right)\left(2-\sqrt{3}\right)$ (ii) $\left(5+\sqrt{5}\right)\left(5-\sqrt{5}\right)$

   (iii) $\left(\sqrt{2}+\sqrt{3}\right)\left(\sqrt{3}+\sqrt{8}\right)$ (iv) $\left(\sqrt{7}-\sqrt{3}\right)^2$

5. Rationalise the denominator.

   (i) $\frac{1}{\sqrt{5}}$ (ii) $\frac{2}{\sqrt{6}}$ (iii) $\frac{\sqrt{21}}{\sqrt{3}}$

6. If a and b are two rational numbers then find the value of a and b in following equations

   (i) $\frac{6+\sqrt{3}}{6-\sqrt{3}}=a+b\sqrt{3}$ (ii) $\frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}-\sqrt{3}}=a+b\sqrt{15}$

7. Simplify

   (i) $\frac{5+\sqrt{3}}{2-\sqrt{3}}+\frac{2-\sqrt{3}}{2+\sqrt{3}}$ (ii) $\frac{2+\sqrt{3}}{2-\sqrt{3}}+\frac{2-\sqrt{3}}{2+\sqrt{3}}+\frac{\sqrt{3}-1}{\sqrt{3}+1}$

8. If $x=3-2\sqrt{2}$ then find the value of $x+\frac{1}{x}$.

What Have We Learnt



1. Any number is said to be a rational number, if it can be written in form $\frac{p}{q}$, $(q \neq 0)$ where p and q are integers.
2. Any number is said to be an irrational number, if it can not be written in form $\frac{p}{q}(q \neq 0)$ when p and q are integers.
3. To find a rational number between two rational numbers, take average of both rational numbers.
4. There are so many rational numbers lying between two rational numbers that they are not countable.
5. Decimal expansion of a rational number is either terminating or non terminating and recurring.
6. Decimal expansion of irrational number is non terminating and non-recurring.
7. $\sqrt{2}, \sqrt{3}, \sqrt{5}, \pi$ are irrational numbers.
8. We get a set of real numbers by combining all rational and irrational numbers.
9. Every point on the number line corrosponds to a unique real number and every real number corresponds to a unique point on the number line.
10. By adding, subtracting, multiplying or dividing a rational and an irrational number gives us an irrational number.
11. To rationalization of the denominator of $\frac{1}{\sqrt{a} - b}$, We multiply it by $\frac{\sqrt{a}+b}{\sqrt{a}+b}$ when a and b are integers.

